Setup OCUDU Testbed

Following the docker based setup available at https://github.com/excellencior/ocudu-testbed-setup

gNB deployment:

Changes for the setup:

(a) **Which RF frontend to use:** ZeroMQ (alternative to physical radio)
Reference: https://docs.ocudu.org/tutorials/srsue/#gnb-2 but with a slight change in the configurations ⇒ adopted configurations from **Split CU and DU** setup:
https://docs.ocudu.org/tutorials/cu_du_split/#ocudu-du-2
Diff details:

ocudu > configs > ! gnb_rf_zmq_srsUE.yml

```
cu_cp:
  amf:
-   addr: 10.53.1.2                 # The address or hostname of the AMF. Check C
    port: 38412
-   bind_addr: 10.53.1.1            # A local IP that the gNB binds to for traffi
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1
-   inactivity_timer: 7200          # Sets the UE/PDU Session/DRB inactivity time

ru_sdr:
-   device_driver: zmq
-   device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_sra
-   srate: 11.52
-   tx_gain: 0
-   rx_gain: 0

cell_cfg:
-   dl_arfcn: 368500                # ARFCN of the downlink carrier (center frequ
-   band: 3                         # The NR band.
-   channel_bandwidth_MHz: 20       # Bandwith in MHz. Number of PRBs will be aut
-   common_scs: 15                  # Subcarrier spacing in kHz used for data.
-   plmn: "00101"                   # PLMN broadcasted by the gNB.
-   tac: 7                          # Tracking area code (needs to match the core
    pdcch:
      common:
        ss0_index: 0                # Set search space zero index to match srsUE
-       coreset0_index: 12          # Set search CORESET Zero index to match srsU
      dedicated:
        ss2_type: common            # Search Space type, has to be set to common
        dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
    prach:
      prach_config_index: 1         # Sets PRACH config to match what is expected
```

```
cu_cp:
  amf:
+   addrs: 10.53.1.2
    port: 38412
+   bind_addrs: 10.53.1.1
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1

ru_sdr:
+   device_driver: zmq
+   device_args: tx_port=tcp://*:2000,rx_port=tcp://10.53.1.4:2001,base_srate=11.52e6
+   srate: 11.52                    # Sample rate in MHz, must match base_srate above
+   tx_gain: 0                      # Gain is not used in ZMQ mode
+   rx_gain: 0                      # Gain is not used in ZMQ mode

cell_cfg:
+   dl_arfcn: 368500                # DL ARFCN corresponding to band 3
+   band: 3                         # FDD band 3, set to match srsUE capabilities
+   channel_bandwidth_MHz: 10       # 10 MHz cell bandwidth, set to match srsUE capabil
+   common_scs: 15                  # 15 kHz subcarrier spacing, set to match srsUE ca
+   plmn: "00101"                   # PLMN, must match the core network and UE subscrit
+   tac: 7                          # Tracking area code, must match the CU-CP configu
    pdcch:
      common:
        ss0_index: 0                # Set search space zero index to match srsUE capal
+       coreset0_index: 6           # Set search CORESET Zero index to match srsUE ca
      dedicated:
        ss2_type: common            # Search Space type, has to be set to common
        dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
    prach:
      prach_config_index: 1         # Sets PRACH config to match what is expected by s
+     total_nof_ra_preambles: 64    # Sets number of available PRACH preambles
+     nof_ssb_per_ro: 1             # Sets the number of SSBs per RACH occasion.
+     nof_cb_preambles_per_ssb: 64  # Sets the number of contention based preambles pe
+   pdsch:
+     mcs_table: qam64              # Sets PDSCH MCS to 64 QAM
+   pusch:
+     mcs_table: qam64              # Sets PUSCH MCS to 64 QAM
+
+ log:
+   filename: /tmp/gnb.log
+   all_level: info
```

(b) **Configuring gNB for ZeroMQ**
After setting up the config-yml for gNB, I had to change a bit in the **docker-compose.yml**, **Dockerfile** for the gNB build
Diff details:
https://github.com/excellencior/ocudu-testbed-setup/commit/a119d335a66512d632937ecf8f91be7eda6166cd
https://github.com/excellencior/ocudu-testbed-setup/commit/e9d5d264d01c1d833c28768e9c1a6918fe9bc6e8

Issues and Fixes (referring to the diffs provided just above)

(a) (Dockerfile) At build time, the rohc-lib.org was unavailable so I pulled from a 3rd party github repo **rohc-2.3.1.taz.xz** (https://github.com/excellencior/ocudu-testbed-setup/blob/a119d335a66512d632937ecf8f91be7eda6166cd/ocudu/docker/scripts/build_rohc.sh#L26)

(b) Found out that using **EXTRA_CMAKE_ARGS** in docker-compose.yml I can provide a switch to enable ZMQ build with gNB. But later I found out in the CMake recipe keeps this flag **on by default** (https://github.com/ocudu/ocudu/blob/9861d780d1fd25ae546739fdcbbb98dbfc4376bd/CMakeLists.txt#L66)

But still there's an issue:

```
apurboturjo  ~/My/OCUDU/ocudu/docker  setup/ocudu  1  4  ? 2  ↑0  ↓0  +9 -36
❯ docker logs ocudu_gnb

---- OCUDU gNB (commit ) ----

Lower PHY in executor sequential baseband mode.
Available radio types: difi and realtime_loopback.
Failed to load RF plugin libocudu_radio_zmq.so: libocudu_radio_zmq.so: cannot open shared object file: No such file or directory
Factory for radio type zmq not found. Make sure to select a valid type.
OCUDU ERROR: Unable to create radio session.
```

My fix:

Explicitly installing the required libraries for ZMQ during multi-phase build in the Dockerfile

builder-base-common (phase)

ocudu/docker/Dockerfile

```
37  37
38  38    COPY docker/scripts/ /src/docker/scripts/
39  39
40  40    RUN --mount=type=secret,id=rh_entitlement_cert,target=/etc/pki/entitlement/entitlement.pem \
41  41        --mount=type=secret,id=rh_entitlement_key,target=/etc/pki/entitlement/entitlement-key.pem \
42  42        --mount=type=secret,id=rh_rhsm_conf,target=/etc/rhsm/rhsm.conf \
43  43        --mount=type=secret,id=rh_rhsm_ca,target=/etc/rhsm/ca/redhat-uep.pem \
44      -     /src/docker/scripts/install_dependencies.sh all && \
    44  +     /src/docker/scripts/install_dependencies.sh build && \
45  45        /src/docker/scripts/install_docker_dependencies.sh build
46  46
    47  + RUN apt-get update && \
    48  +     apt-get install -y libzmq3-dev && \
    49  +     rm -rf /var/lib/apt/lists/*
    50  +
47  51    ################################
48  52    # Stage 0b: Shared lean base   #
49  53    ################################
```

I tried setting the install_dependencies.sh to use arg **all** but it still failed during the runtime-base phase. So, I decided to go with explicit installation afterwards.

runtime-base (phase)

ocudu/docker/Dockerfile

```
155 155   ARG COMPONENT="ocu ocucp ocuup odu odu_low odu_split_8 odu_split_7_2 gnb gnb_split_8 gnb_split_7_2 ru_emulator"
156     - ARG ENABLE_UHD=On
157     - ARG ENABLE_DPDK=On
    156 + ARG ENABLE_UHD=Off
    157 + ARG ENABLE_DPDK=Off
158 158   ARG STRIP_BIN=0
159 159
160 160   ENV ROHC_DIR=/opt/rohc
        @@ -383,6 +383,10 @@ RUN --mount=type=secret,id=rh_entitlement_cert,target=/etc/pki/entitlement/entit
383 383       /usr/local/etc/install_rohc_dependencies.sh run && \
384 384       /usr/local/etc/install_docker_dependencies.sh run
385 385
    386 + RUN apt-get update && \
    387 +     apt-get install -y libzmq5 && \
    388 +     rm -rf /var/lib/apt/lists/*
    389 +
386 390   # Register DPDK/UHD/ROHC library paths for the dynamic linker.
387 391   # When binaries have file capabilities, ld.so enters secure-execution mode and
```

Final outcome:

root@58d9b829b4a3: / — docker exec -it ocudu_gnb bash

```
root@58d9b829b4a3:/# cat /tmp/gnb.log | grep "zmq" | head -n 20
  device_driver: zmq
2026-11-07T18:47:48.788111 [zmq:tx:0:0] [I] Binding to address tcp://*:2000.
2026-11-07T18:47:48.816228 [zmq:rx:0:0] [I] Connecting to address tcp://10.53.1.4:2001.
2026-11-07T18:47:50.919800 [zmq:rx:0:0] [I] Waiting for data.
2026-11-07T18:47:50.919843 [zmq:tx:0:0] [I] Waiting for request.
2026-11-07T18:47:50.919984 [zmq:rx:0:0] [I] Waiting for reading samples. Completed 0 of 11520 samples.
2026-11-07T18:47:51.919802 [zmq:rx:0:0] [I] Waiting for data.
2026-11-07T18:47:51.919843 [zmq:tx:0:0] [I] Waiting for request.
2026-11-07T18:47:51.920024 [zmq:rx:0:0] [I] Waiting for reading samples. Completed 0 of 11520 samples.
2026-11-07T18:47:52.919802 [zmq:rx:0:0] [I] Waiting for data.
2026-11-07T18:47:52.919844 [zmq:tx:0:0] [I] Waiting for request.
2026-11-07T18:47:52.920077 [zmq:rx:0:0] [I] Waiting for reading samples. Completed 0 of 11520 samples.
2026-11-07T18:47:53.919802 [zmq:rx:0:0] [I] Waiting for data.
2026-11-07T18:47:53.919844 [zmq:tx:0:0] [I] Waiting for request.
2026-11-07T18:47:53.920093 [zmq:rx:0:0] [I] Waiting for reading samples. Completed 0 of 11520 samples.
2026-11-07T18:47:54.919804 [zmq:rx:0:0] [I] Waiting for data.
```

5gc deployment:

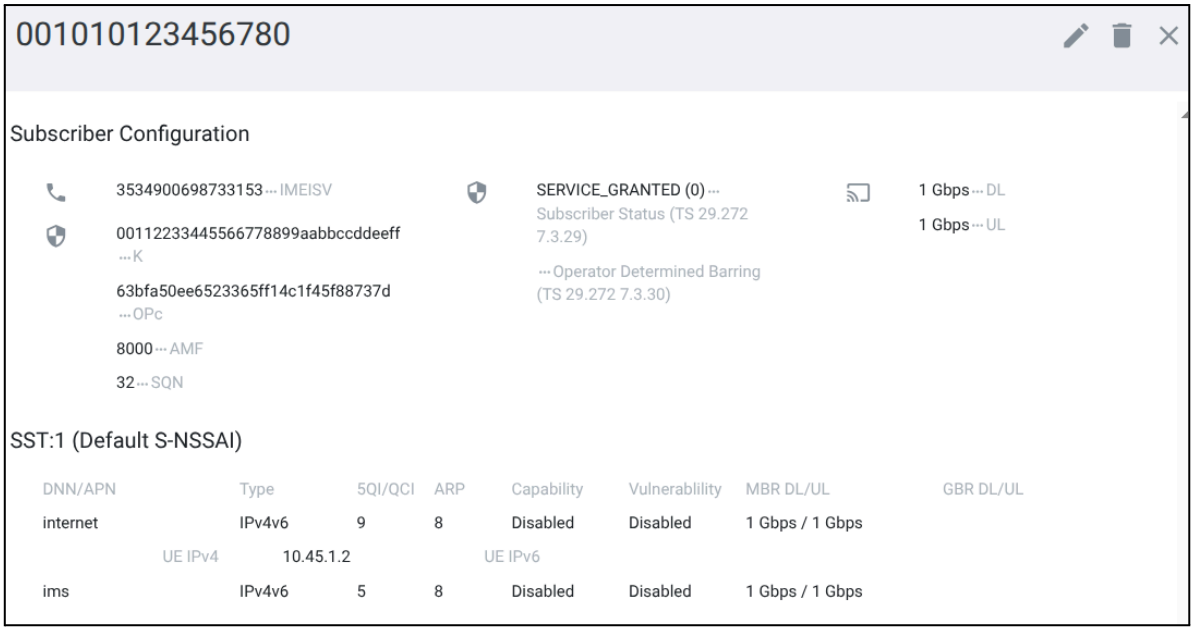
As for the 5g-core deployment I didn't have to change much but got an overall idea of how it works:

(a) The NFs' identities and configurations are available at **ocudu/docker/open5gs/open5gs-5gc.yml**

(b) It in the current configuration supports a single UE based deployments loaded from **ocudu/docker/open5gs/open5gs.env** but can be converted into multi-ue setup by provisioning the intended UE configurations mainly (K, OPc, imsi) via **subscribers.csv** and updating the .env to use the csv. ⇒ add_users.py



The ue_conf provided data matches with the one to be provisioned into the embedded mongodb at 5gc



MongoDB

```
test> printjson(db.getSiblingDB("open5gs").subscribers.find({}, { imsi:1, "security.k":1, "security.opc":1, "slice.session.ue.ipv4":1 }))
[
  {
    _id: ObjectId('6aed8499a1aa1de13791e4c7'),
    imsi: '001010123456780',
    slice: [
      {
        session: [
          {
            ue: {
              ipv4: '10.45.1.2'
            }
          },
          {}
        ]
      }
    ],
    security: {
      k: '00112233445566778899aabbccddeeff',
      opc: '63bfa50ee6523365ff14c1f45f88737d'
    }
  }
]
```

Finally, I tried to test the NAS Registration, but I noticed that there's an issue.

ASN1 [E] Decoding Failed (5GS tracking area identity list): Length (19) does not equal expected length 7

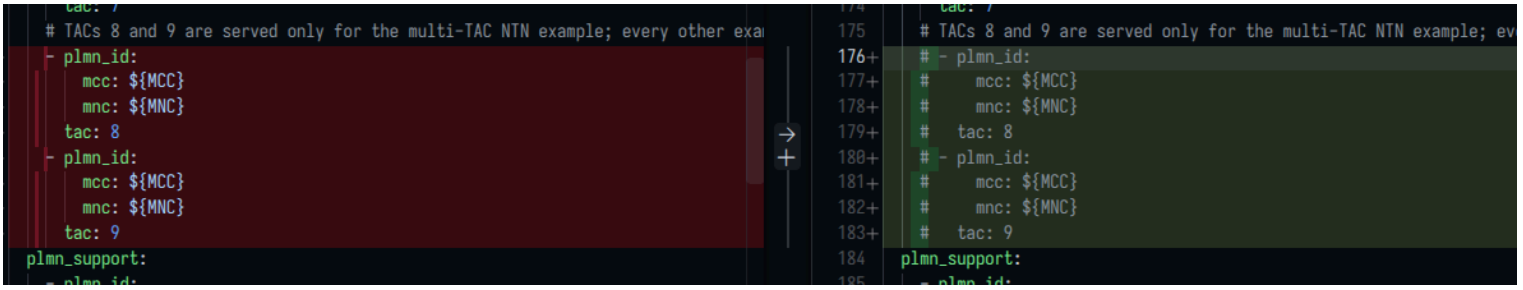
```
2026-11-08T19:21:20.375777 [ASN1    ] [E] [    0] Decoding Failed (5GS tracking area identity list): Length (19) does
not equal expected length 7
2026-11-08T19:21:20.375779 [ASN1    ] [E] [    0] [/srsRAN_4G/lib/src/asn1/nas_5g_msg.cc][507] Decoding failure.
2026-11-08T19:21:20.375780 [ASN1    ] [E] [    0] [/srsRAN_4G/lib/src/asn1/nas_5g_msg.cc][3912] Decoding failure.
2026-11-08T19:21:20.375780 [ASN1    ] [E] [    0] [/srsRAN_4G/lib/src/asn1/nas_5g_msg.cc][3888] Decoding failure.
```

From the log I was able to find the related code segment at: **/srsRAN_4G/lib/src/asn1/nas_5g_ies.cc**

```
// IE: 5GS tracking area identity list
// Reference: 9.11.3.9
SRSASN_CODE tracking_area_identity_list_5gs_t::unpack(asn1::cbit_ref& bref)
{
  uint8_t length = 0;
  HANDLE_CODE(bref.unpack(length, 8));
  if (length != 7) {
    asn1::log_error("Decoding Failed (5GS tracking area identity list): Length (%d) does not equal expected length 7",
                    length);
    return asn1::SRSASN_ERROR_DECODE_FAIL;
  }

  // TODO proper unpacking
  bref.advance_bits(length * 8);
  return SRSASN_SUCCESS;
}
```

Then after some analysis and digging I came to find out that the 5gc.amf advertised 3 TACs (7, 8, 9) in the RA message. But srsUE's NAS decoder expected a single-TAC (7 bytes). So, srsUE received a multi-TAC list (19 bytes)



So, commenting out these blocks and then redeploying 5gc followed by gnb and srsUE solved the issue.

srsUE deployment

I used the srsRAN-4G project as mentioned in the documentation but for the RF medium used ZeroMQ
Reference: https://docs.ocudu.org/tutorials/srsue/

I had two options for the setup. One is setting up the project in the host machine and another is trying dockerizing it to keep it as a separate container consistent with the whole setup.

Dockerization

Issues faced:

Packing the project inside of the docker starting with base image and installing required dependencies as mentioned in the documentation.

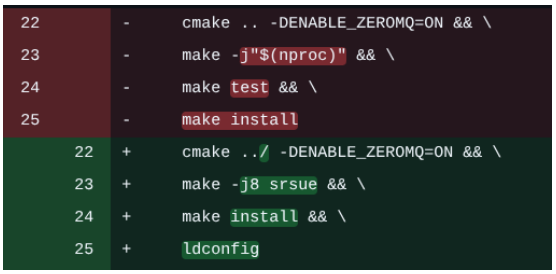
(1) Many build failures originated from compiling and running tests of all the modules from the project. I got a hint from the error log:

```
Shell
#11 216.9 [ 30%] Building CXX object
srsenb/src/stack/mac/CMakeFiles/srsenb_mac.dir/sched_ue_ctrl/sched_ue_cell.cc.o
#11 217.0 [ 30%] Building C object
lib/src/phy/fec/block/test/CMakeFiles/block_test.dir/block_test.c.o
#11 217.1 [ 30%] Building C object
lib/src/phy/fec/convolutional/test/CMakeFiles/viterbi_test.dir/viterbi_test.c.o
#11 217.2 /srsRAN_4G/lib/src/phy/fec/block/test/block_test.c: In function 'test':
#11 217.2 /srsRAN_4G/lib/src/phy/fec/block/test/block_test.c:62:11: error: writing 1 byte into a
region of size 0 [-Werror=stringop-overflow=]
#11 217.2    62 |     tx[i] = (uint8_t)srsran_random_uniform_int_dist(random_gen, 0, 1);
#11 217.2       |     ~~~~~~^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#11 217.2 /srsRAN_4G/lib/src/phy/fec/block/test/block_test.c:54:18: note: at offset 11 into
destination object 'tx' of size 11
#11 217.2    54 |   uint8_t        tx[SRSRAN_FEC_BLOCK_MAX_NOF_BITS]  = {};
#11 217.2       |                  ^~
#11 217.3 cc1: all warnings being treated as errors
#11 217.3 make[2]: *** [lib/src/phy/fec/block/test/CMakeFiles/block_test.dir/build.make:79:
lib/src/phy/fec/block/test/CMakeFiles/block_test.dir/block_test.c.o] Error 1
#11 217.3 make[1]: *** [CMakeFiles/Makefile2:5845:
lib/src/phy/fec/block/test/CMakeFiles/block_test.dir/all] Error 2
#11 217.3 make[1]: *** Waiting for unfinished jobs....
```

`/srsRAN_4G/lib/src/phy/fec/block/test/block_test.c:62:11: error: writing 1 byte into a region of size 0 [-Werror=stringop-overflow=]`

This particular line means it might be compiling something that I might not require to build the **srsue** module. (guess)

Then I tried with providing an argument (guessed from my experience build tools like gradle takes the focused builds by subproject specification) to build only the srsue project and not run the full test suite / build apart from it:



Line no: 23

GitCommit Reference

(2) Even after specifying the project focus build command and setting up the ZMQ flag the build resulted in an error provided below:

```
Shell
Lower PHY in executor sequential baseband mode.
Available radio types: difi and realtime_loopback.
Failed to load RF plugin libocudu_radio_zmq.so: libocudu_radio_zmq.so: cannot open shared object
file: No such file or directory
Factory for radio type zmq not found. Make sure to select a valid type.
OCUDU ERROR: Unable to create radio session.
```

I took the help of chatgpt to help me resolve this issue:

a. Tried with several combinations of configurations f.e. omitting the **make install** and in the docker-compose **command block**, execute the built binary (from the **/srsRAN_4G/build/srsue/src/srsue**) directly. But it didn't solve the issue.
b. Then it suggested that I should use ldconfig. I noticed it's also being used in the OCUDU project too. It's responsible for updating the system's shared lib-cache. /usr/local/lib/libsrsran_rf.so.0 was located at a proper location and I had to specify ldconfig to update the container's system cache. Which eventually resulted into a working build (provided below). A successful handshake with the already gNB was made.

```
2026-11-08T19:23:30.171376 [RRC-NR ] [D] SRB1 - Rx ueCapabilityEnquiry (8 B)
2026-11-08T19:23:30.171380 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.171398 [RRC-NR ] [I] Received UE Capability Enquiry
2026-11-08T19:23:30.171511 [RRC-NR ] [D] SRB1 - Tx ueCapabilityInformation (30 B)
2026-11-08T19:23:30.171513 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.218528 [RRC-NR ] [D] RX PDU, LCID: 1
2026-11-08T19:23:30.218594 [RRC-NR ] [D] SRB1 - Rx dlInformationTransfer (49 B)
2026-11-08T19:23:30.218598 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.218645 [RRC-NR ] [D] Preparing UL Info Transfer
2026-11-08T19:23:30.218669 [RRC-NR ] [D] SRB1 - Tx ulInformationTransfer (13 B)
2026-11-08T19:23:30.218670 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.218722 [RRC-NR ] [D] Preparing UL Info Transfer
2026-11-08T19:23:30.218741 [RRC-NR ] [D] SRB1 - Tx ulInformationTransfer (38 B)
2026-11-08T19:23:30.218743 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.345707 [RRC-NR ] [D] RX PDU, LCID: 1
2026-11-08T19:23:30.345783 [RRC-NR ] [D] SRB1 - Rx dlInformationTransfer (52 B)
2026-11-08T19:23:30.345794 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.345850 [RRC-NR ] [D] Preparing UL Info Transfer
2026-11-08T19:23:30.345893 [RRC-NR ] [D] SRB1 - Tx ulInformationTransfer (13 B)
2026-11-08T19:23:30.345894 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.412075 [RRC-NR ] [D] RX PDU, LCID: 1
2026-11-08T19:23:30.412201 [RRC-NR ] [D] SRB1 - Rx rrcReconfiguration (110 B)
2026-11-08T19:23:30.412211 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.412260 [RRC-NR ] [I] Proc "NR Connection Reconfiguration" - Starting...
2026-11-08T19:23:30.412391 [RRC-NR ] [D] Proc "NR Connection Reconfiguration" - RRC NR Reconfiguration: {
2026-11-08T19:23:30.412505 [RRC-NR ] [D] Proc "NR Connection Reconfiguration" - Master Cell Group: {
2026-11-08T19:23:30.412506 [RRC-NR ] [I] Proc "NR Connection Reconfiguration" - Applying Master Cell Group Cfg.
2026-11-08T19:23:30.412630 [RRC-NR ] [I] Adding lcid 4 and radio bearer ID 1
2026-11-08T19:23:30.413325 [RRC-NR ] [I] Proc "NR Connection Reconfiguration" - Applying Radio Bearer Cfg.
2026-11-08T19:23:30.413326 [RRC-NR ] [D] Applying SRB Add/Mod to SRB2
2026-11-08T19:23:30.413342 [RRC-NR ] [D] Applying DRB Add/Mod to DRB1
2026-11-08T19:23:30.413371 [RRC-NR ] [D] No Security Config Present
2026-11-08T19:23:30.413372 [RRC-NR ] [D] Preparing RRC Connection Reconfig Complete
2026-11-08T19:23:30.413392 [RRC-NR ] [D] SRB1 - Tx rrcReconfigurationComplete (2 B)
2026-11-08T19:23:30.413394 [RRC-NR ] [D] Content:[
2026-11-08T19:23:30.416701 [RRC-NR ] [I] Proc "NR Connection Reconfiguration" - Finished NR Connection Reconfiguration successfully
```

SIB1 received from gNB, then SRB0 for the setup request to the gNB
Finally, SRB0 (Rx) rrcSetup received from gNB, SRB1 (Tx) rrcSetupComplete to gNB followed by Connection Setup Success log.

As for the configuration file **ue_rf.conf**:
https://github.com/excellencior/ocudu-testbed-setup/blob/setup/ocudu/ocudu/docker/srsue/ue_rf.conf
It was referenced from: https://docs.ocudu.org/tutorials/srsue/#srsue

NETWORK CONNECTION

```
root@srsue:/srsRAN_4G# ip route
default via 10.53.1.1 dev eth0
10.45.1.0/24 dev tun_srsue proto kernel scope link src 10.45.1.2
10.53.1.0/24 dev eth0 proto kernel scope link src 10.53.1.4
root@srsue:/srsRAN_4G# ping -c 5 10.45.1.2
PING 10.45.1.2 (10.45.1.2) 56(84) bytes of data.
64 bytes from 10.45.1.2: icmp_seq=1 ttl=64 time=0.088 ms
64 bytes from 10.45.1.2: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 10.45.1.2: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 10.45.1.2: icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from 10.45.1.2: icmp_seq=5 ttl=64 time=0.062 ms

--- 10.45.1.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4077ms
rtt min/avg/max/mdev = 0.058/0.067/0.088/0.010 ms
```

The network tunnel was deployed properly and the srsUE was able to connect to the 5gc

UE ⇒ 5gc

```
root@srsue:/srsRAN_4G# cat /tmp/ue.log | grep -i "registration"
2026-11-08T19:23:29.049864 [NAS5G  ] [I] Sending Registration Request
2026-11-08T19:23:30.218633 [NAS5G  ] [I] Handling Registration Accept
2026-11-08T19:23:30.218643 [NAS5G  ] [I] Sending Registration Complete
```

5gc

```
❯ docker logs open5gs_5gc | grep -i "registration"
(node:95) DeprecationWarning: collection.ensureIndex is deprecated. Use createIndexes instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:95) DeprecationWarning: collection.count is deprecated, and will be removed in a future version. Use Collection.countDocuments or Collection.estimatedDocumentCount ins
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Open5GS daemon v2.7.6

11/08 20:23:06.304: [app] INFO: Configuration: 'open5gs-5gc.yml' (../lib/app/ogs-init.c:144)
11/08 20:23:29.974: [gmm] INFO: Registration request (../src/amf/gmm-sm.c:1339)
11/08 20:23:30.334: [gmm] INFO: [imsi-001010123456780] Registration complete (../src/amf/gmm-sm.c:2698)
```